When and How Might You Die?

Visualizing Age & Cause of Death using UN & WHO Data

Link to Viz: https://observablehq.com/d/f63cf70b0a1eee0e

Problem & Motivation

“When and how will I die?”

Problem with life expectancy & general cause-of-death stats:

- Feels too distant
- Does not show the full picture of risks over lifetime

Life Expectancy

Print

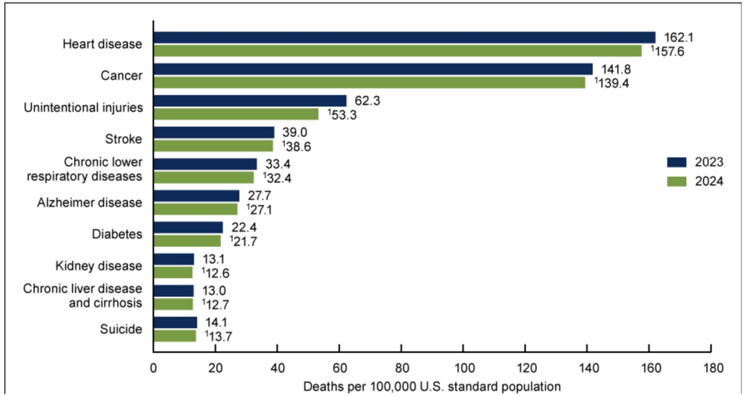
Data are for the U.S.

Life expectancy at birth

- Both sexes: 79.0 years
- Males: 76.5 years
- Females: 81.4 years

Source

Figure 4. Age-adjusted death rate for the 10 leading causes of death in 2024: United States, 2023 and 2024



Source

Problem & Motivation

“When and how will I die?”

Goals:

- Turn population data into individual experience
- Show risks individuals experience across different stages of life

Problem & Motivation

"When and how will I die?"

Proposed Approach:

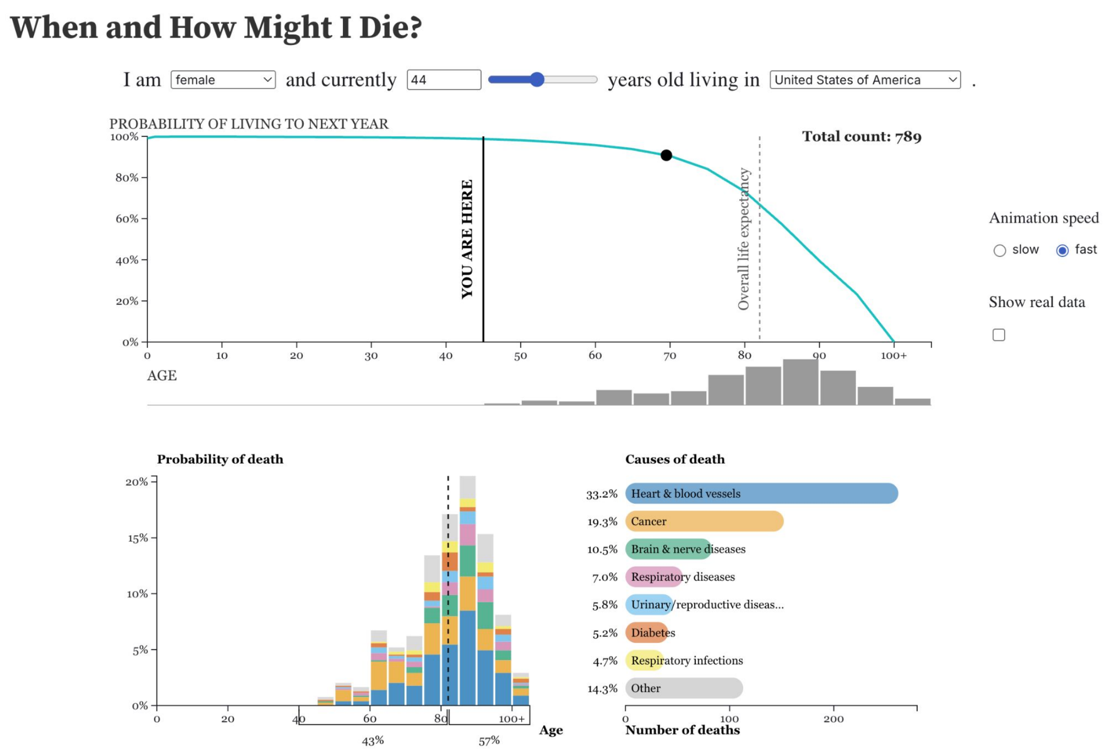
- Use population data to simulate when and how people die
  - Inspired by https://flowingdata.com/2015/09/23/years-you-have-left-to-live-probably/
- Visualize both the process and results

Datasets

- United Nations (UN) mortality statistics
  - https://population.un.org/dataportal/home?df=9456c5d3-9e5a-409a-a65d-1703294f95bb
- World Health Organization (WHO) causes of death data
  - https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghe-leading-causes-of-death

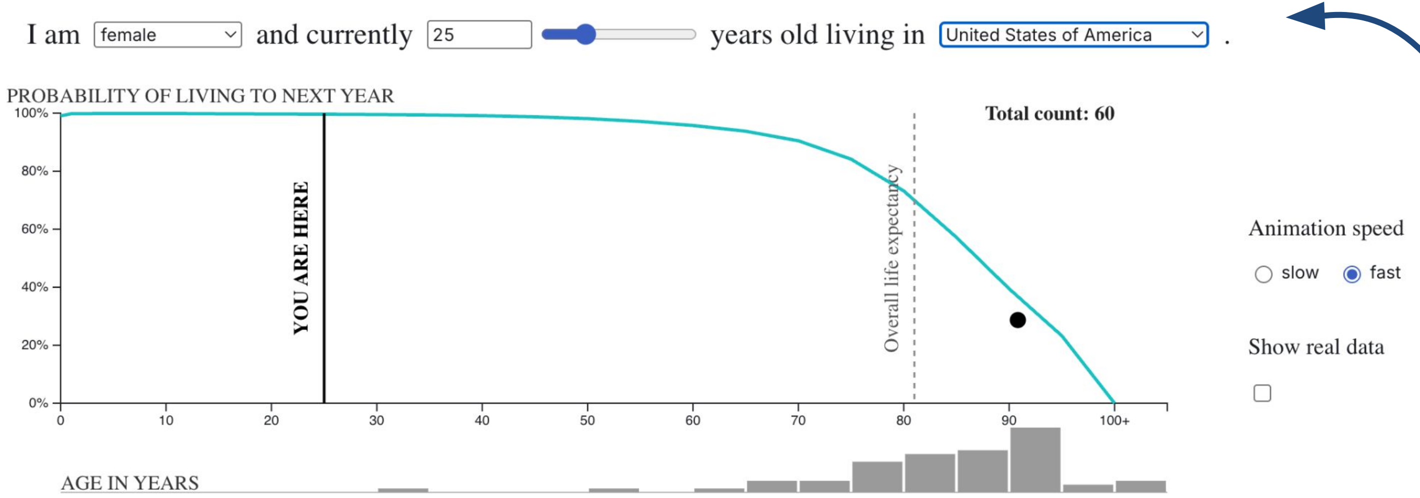
Key Visualization

Overall View



Key Visualization

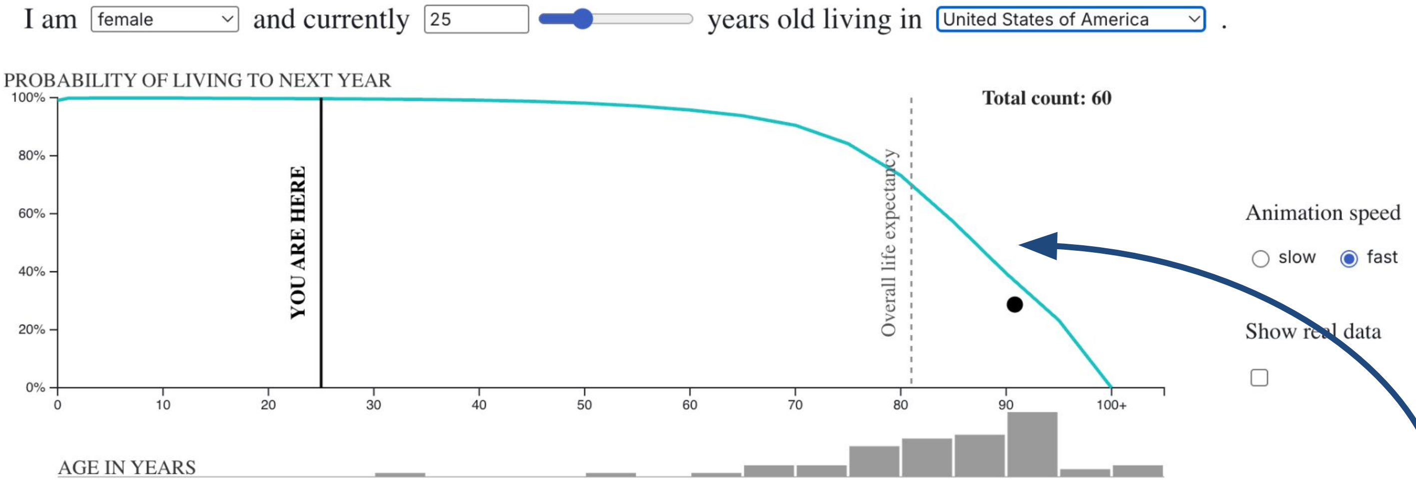
User selection



Users select their gender, current age, and country

Key Visualization

Survival Probability Chart



Green curve shows corresponding survival probability across ages

Key Visualization

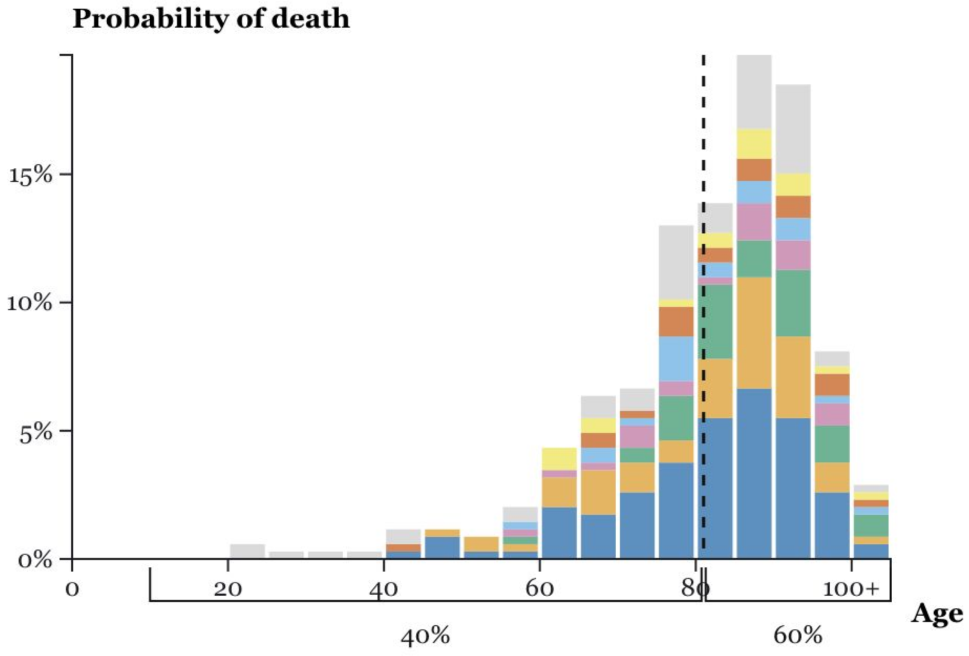
Animated simulation

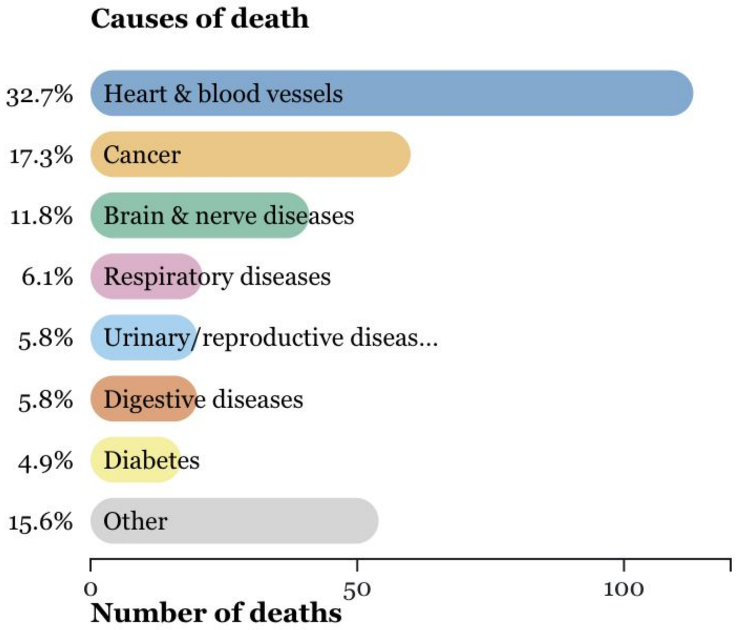
See demo video

- A moving dot represents one simulated life
- Starts at the current age
- Moves along the survival probability curve
- At each age, death is determined by the probability of survival
- When death occurs, the dot falls
- After death:
  - A cause of death is assigned
  - Based on gender, region, and age
- A new dot appears and the simulation repeats

Key Visualization

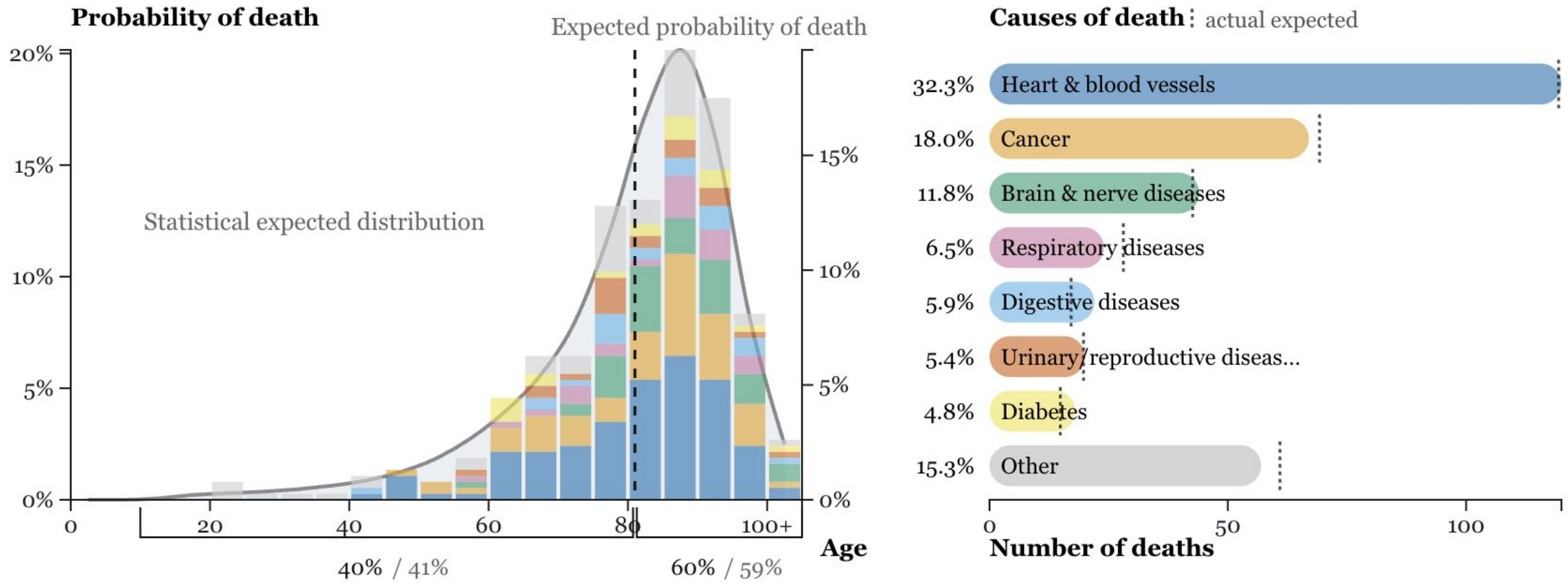
Death age distribution & Cause-of-death counts





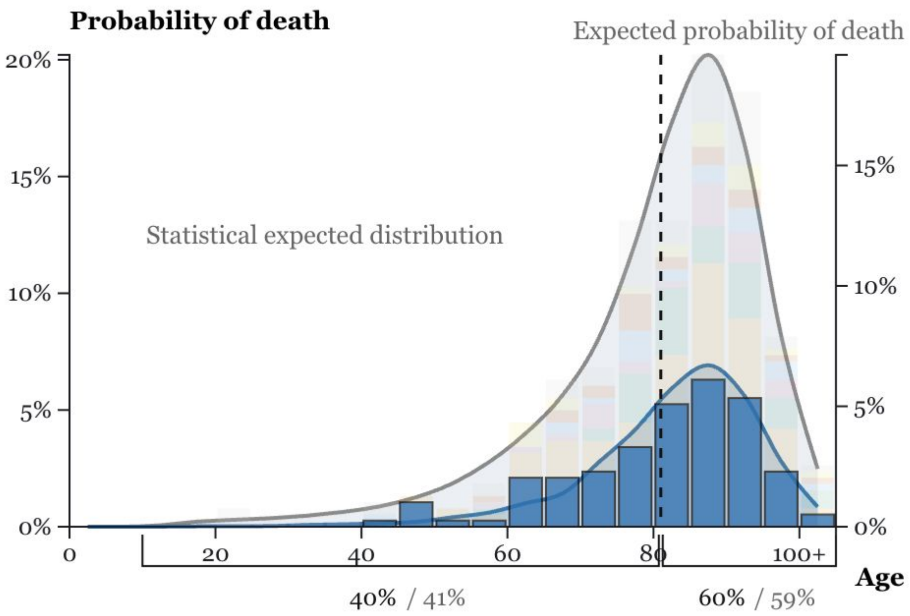
Key Visualization

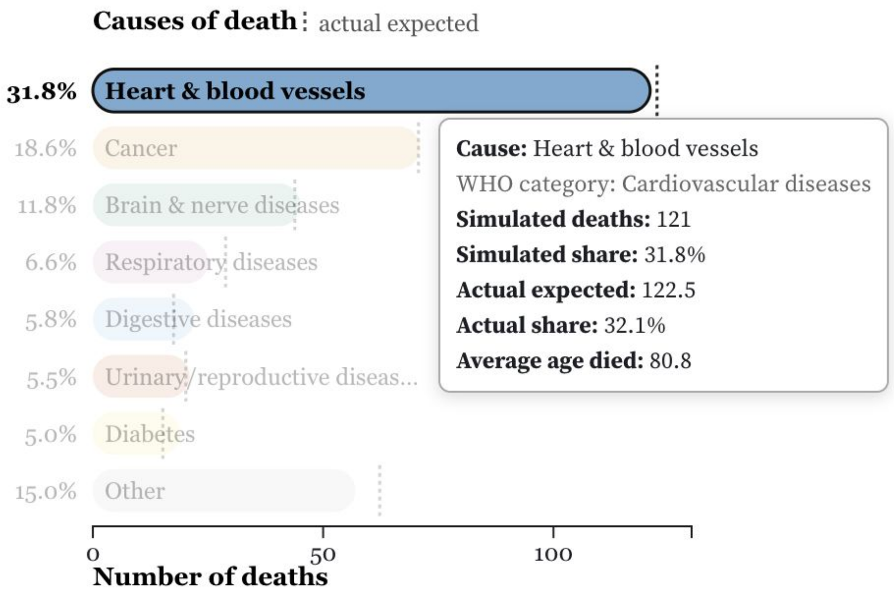
Display real data on top of simulated results



Key Visualization

Highlight & Hover (example)

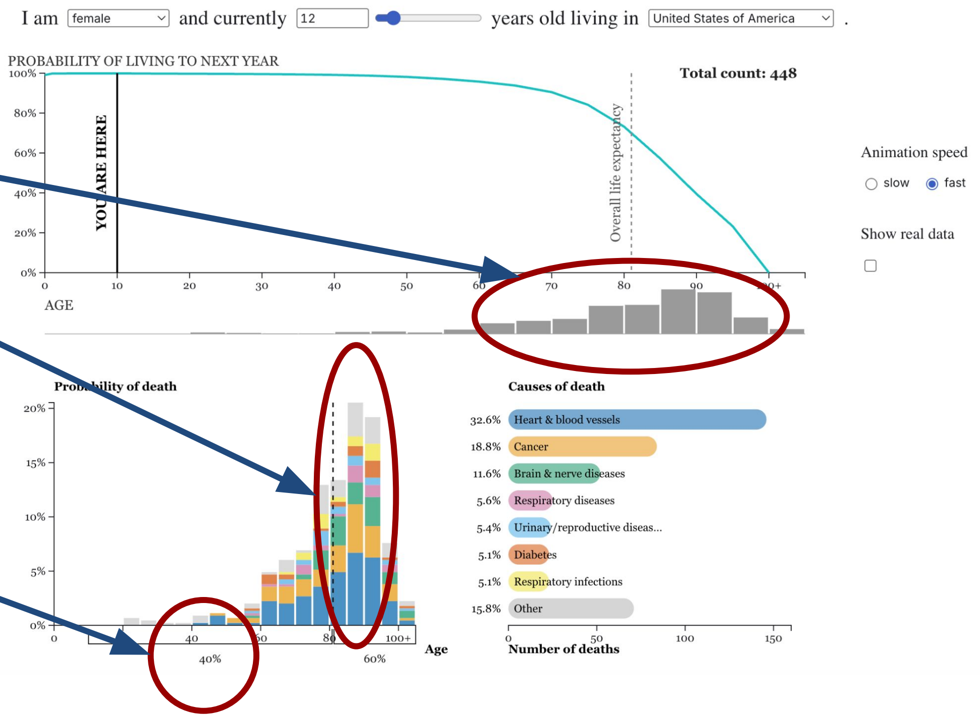




Insights

Life expectancy is only a reference point

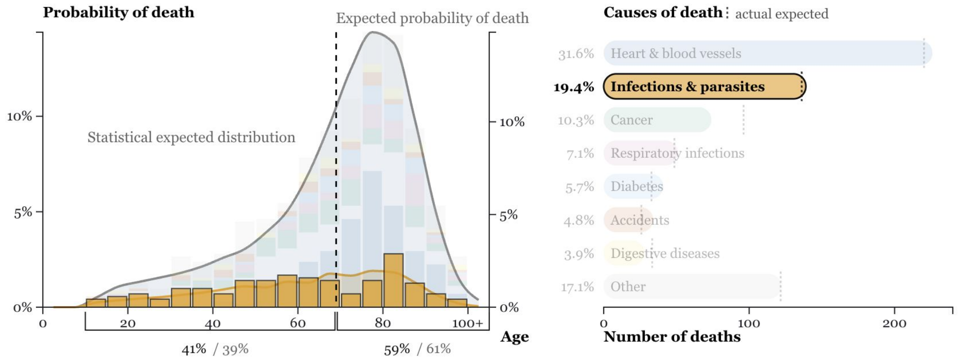
- Possible ages of death are spread over a wide range
- Highest probability of death may occur after life expectancy
- High probability of not reaching life expectancy



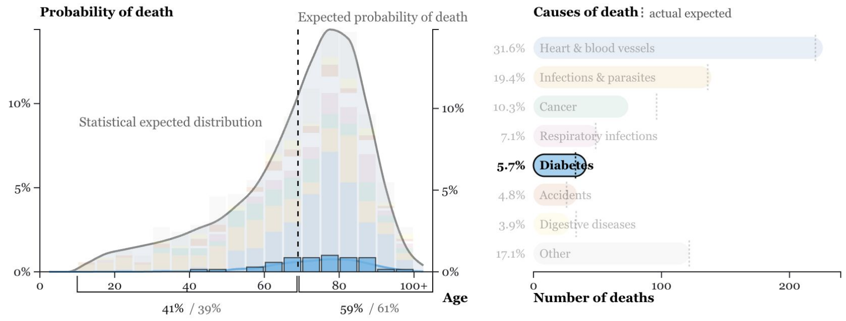
Insights

Causes of death differ across life stages

- Some causes appear across almost all ages

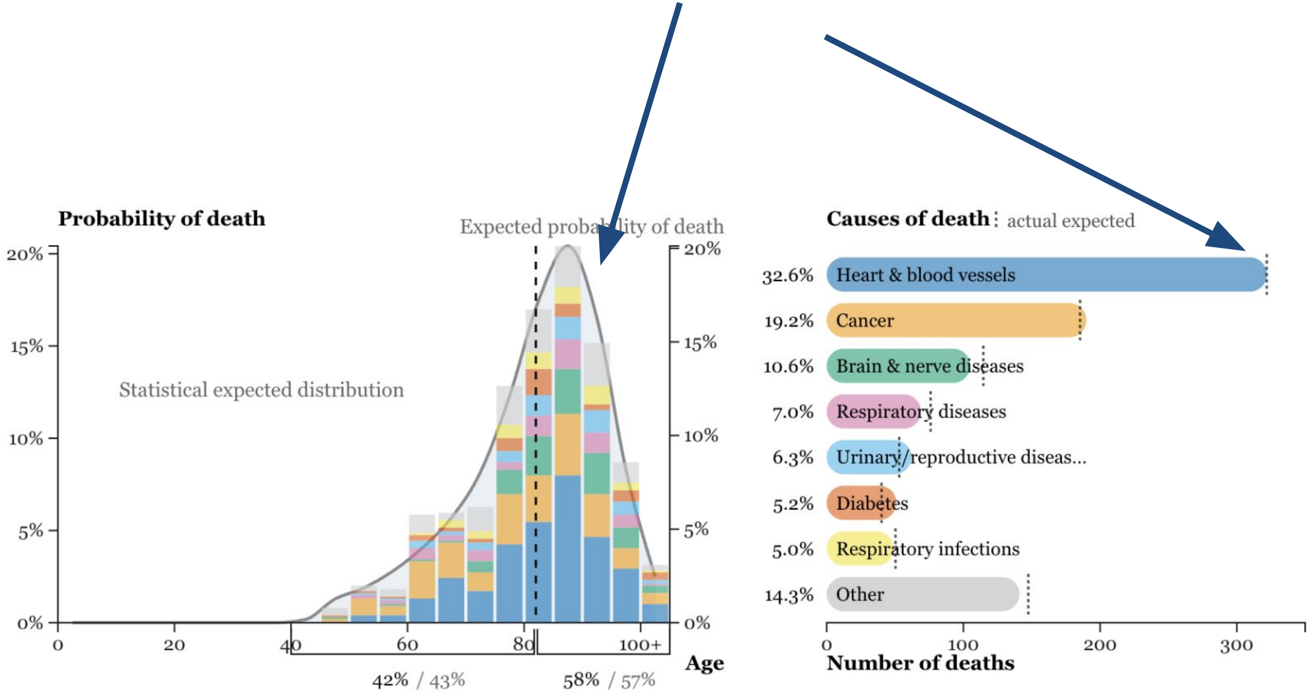


- Others are concentrated in specific stages of life.



Insights

Individual randomness accumulates into population patterns



Limitations & future work

- Not practical to display causes of death clearly
  - Future: allow expanding or drilling down into "Other"

- Results are not truly country-specific
  - WHO data only available for 5 aggregate regions. Selected countries mapped to those regions
  - Future: use country-level data if available

Limitations & future work

- Had to use 2019 cause-of-death data
  - Latest WHO data heavily influenced by COVID
  - Future: use post-COVID data when available for more accurate patterns

Contributions

- Connects population data to individual experience
- Makes abstract data intuitive

- Helps understand uncertainty in ages and causes of death
- Helps understand distribution of ages and causes of death
- Shows how randomness forms stable patterns